

Calibrated Conviction: A Multi-Source Ensemble that Knows When Not to Predict in the Indian Equity Market

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Abstract

Many published equity-prediction systems advertise directional accuracies that collapse once their probabilities are calibrated and realistic transaction costs are charged. We went the other way, asking what a forecaster in an emerging market can be trusted to deliver. Eight years of National Stock Exchange of India data give a blunt answer: next-period single-name direction is essentially unpredictable, and a cross-sectional gradient-boosted model confirms it at 50.1% out-of-sample accuracy—no better than the upward drift. The value lies elsewhere: a three-stage procedure cuts the hybrid ensemble’s expected calibration error from 0.35 to 0.05, and a conviction-gated swing signal, at a thirty-trading-day horizon, lifts the realized hit-rate to 60.6% against a 58.1% base rate while firing on only 5.7% of observations and beating it in four of five test years. Every figure here is a measured walk-forward statistic.

Keywords: Equity forecasting; Probability calibration; Conformal prediction; Ensemble

learning; Conviction gating

Highlights

- Daily single-name NSE direction carries no edge beyond drift (50.1% OOS).
- A three-stage calibration cuts expected calibration error from 0.35 to 0.05.
- A conviction gate lifts the 30-day hit-rate to 60.6% vs a 58.1% base rate.
- The high-conviction bucket beats the base rate in four of five test years.
- Every displayed statistic is a measured walk-forward result, not a fit.

1. Introduction

The trouble with much of the machine-learning-for-finance literature is not that the models fail. It is that they seem to succeed. Directional accuracies of 60–70% turn up regularly, but they tend not to survive a few basic checks: are the reported probabilities calibrated, does the evaluation respect the order of time, and are realistic Indian transaction costs charged on every round trip [1]? We applied those checks to our own platform, and the comfortable numbers evaporated. What follows is an account of what was left once they did, and of the engineering that made the remainder worth trusting.

Our setting is the National Stock Exchange of India (NSE). It is a market driven by Foreign and Domestic Institutional Investor (FII/DII) flows and by a large retail base that reacts to news in both English and Hindi. Earlier multi-source studies report that fusing price, news and order-flow can raise accuracy above single-source baselines [2, 3, 4], and hybrid sentiment-indicator architectures have been proposed for movement prediction [5, 6]. What that body of work rarely asks is whether the probabilities it produces can be believed. The efficient-market hypothesis [7] says that short-horizon direction for liquid large-caps ought to be close to unpredictable; our own results agree. For an applied system, then, the job is not to argue the limit away but to work sensibly within it.

We rebuilt the project around four aims: to bring financial metrics, multilingual sentiment, FII/DII flows and macroeconomic state into a single feature space (Section 2.1–2.3); to make the ensemble’s probabilities calibrated rather than merely confident, since a confidence score you cannot trust is worse than none (Section 2.4–2.5); to build a decision layer that stays silent by default and acts only above a conviction threshold validated out of sample, with distribution-free intervals attached (Section 2.6–2.7); and to keep the whole thing explainable and able to audit itself (Section 2.8–2.9). The empirical core in Section 3 is modest about what it claims and open about where it fails.

2. Materials and Methods

2.1 Multi-source data ingestion

Let t index NSE trading days. We align four streams on a common daily grid: adjusted OHLCV bars via `yfinance`; news and social text from NewsAPI, Google News RSS (English and Hindi), the `gnews` library and Economic Times feeds, behind a URL-level domain filter that drops non-financial sources, such as the software registries the word “Reliance” would otherwise match; FII and DII daily net flows in INR crore, lagged one session so nothing leaks from the future; and macroeconomic state—the India VIX, INR/USD, Brent crude and gold as clipped log-returns, with NIFTY-relative strength and beta.

2.2 Feature engineering

Each ticker-day is described by a vector of roughly forty stationary inputs, organized into ten groups: multi-horizon returns (1–20 days); volatility (realized and normalized ATR); momentum and oscillators (Wilder RSI, Bollinger Band %B in Eq. (1), the 12/26/9 MACD line and histogram, Williams %R); volume (volume ratio,

on-balance-volume slope, Chaikin Money Flow); trend strength (Average Directional Index, Aroon, 9/21/50 EMA slopes); lagged divergence flags; chart-pattern distance-to-target; FinBERT sentiment; lagged FII/DII flows; macro returns; and NIFTY-relative strength and beta. Two details guard against leakage: every divergence flag is shifted one period before comparison, so no feature sees its own future, and each column is winsorized and scaled on the training fold alone.

$$\text{RSI}_{14}(t) = 100 - \frac{100}{1 + \text{RS}_{14}(t)}, \quad \text{BB}_{\%B}(t) = \frac{C_t - \text{LB}(t)}{\text{UB}(t) - \text{LB}(t) + \varepsilon} \quad (1)$$

2.3 FinBERT sentiment pipeline

A FinBERT pipeline [8] (the `ProsusAI/finbert` checkpoint) scores each headline and returns a raw polarity $\tilde{s} = p_+ - p_-$. FinBERT tends to over-assign the neutral class on short headlines, so a curated layer of about sixty Indian-market bullish and bearish terms (“upper circuit”, “promoter pledge”, “SEBI ban”) fixes the high-precision cases. Hindi articles are machine-translated first, since the model is English-only. We take the daily ticker score as a source-weighted mean, and a separate large-language-model layer rates novelty and materiality to build a news-alpha term.

2.4 Heterogeneous ensemble and leak-free stacking

We train five base learners per ticker. XGBoost [9] and LightGBM [10] run shallow and L1/L2-regularized for the short per-ticker window; CatBoost [11] joins them; a multi-task LSTM–GRU network carries three heads (return, direction under binary cross-entropy, and a soft-plus volatility head); and ARIMA and Prophet sit in as fading statistical priors. Out-of-fold (OOF) predictions come from a purged, embargoed, expanding-window scheme: a five-day purge and embargo separate each validation block from its training window, so lagged-return features cannot bleed across the boundary—the standard precaution for financial cross-validation [12]. The OOF matrix

then feeds a Ridge meta-learner [13],

$$\hat{s}_t^{\text{stack}} = \beta^\top \mathbf{z}_t + \beta_0, \quad [\beta; \beta_0] = (\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{Z}}^\top \mathbf{y}, \quad (2)$$

and because it is built on out-of-fold rows, leakage has nowhere to enter. A pre-trained cross-sectional LightGBM is let in at a small weight, but only if its own OOF accuracy clears a gate set just above chance.

2.5 Three-stage probability calibration

No single change mattered more. An earlier version of the system was badly miscalibrated: it reported an average up-probability near 0.80 against a realized base rate near 0.48, an expected calibration error (ECE) of roughly 0.35, and a Brier score worse than a constant forecast. The cause was twofold. An isotonic calibrator fitted on the tree-average OOF distribution had been applied to the Ridge meta-output, a different distribution; and the final convex blend of two individually calibrated probabilities was never recalibrated, even though a convex combination of calibrated forecasts is not itself calibrated. The fix has three stages: an isotonic regression [14] fitted on the tree-average OOF predictions and applied to the matching tree-average test predictions; a validation-fitted Platt scaler [15] on the recurrent network’s sigmoid output; and, after the 0.60/0.40 blend, a logistic recalibration on the held-out validation fold that pulls a low-skill model back toward the base rate rather than letting it claim confidence it has not earned. Calibration is measured by ECE over ten bins and the Brier score [16], and the same transform is applied to the live inference row, so the probability a user sees sits on the same corrected scale.

2.6 Conformal prediction intervals

Every point forecast carries a distribution-free interval. Two quantile XGBoost regressors are trained at the 0.10 and 0.90 levels, and the split-conformal half-width is

the finite-sample-corrected quantile of absolute residuals, on a held-out fold disjoint from where the band is applied [17, 18]. Under exchangeability the interval has a marginal coverage guarantee; in practice, serial dependence means the coverage we observe is just that—an observation, not a guarantee.

2.7 Conviction-gated swing-direction signal

Since next-day direction proved unpredictable (Section 3.1), the signal we trade aims at a longer, more tractable question: over the next thirty trading days—roughly six calendar weeks—is a name more likely than not to be higher than today, and is the setup convincing enough to act on? A cross-sectional logistic model takes nine point-in-time features: 12-1 month momentum [19], intermediate-term and quarterly returns, a five-day reversal, distance from the 200-day average, a golden-cross flag, RSI, realized volatility, and an above-average flag. We evaluate it with an embargoed, expanding-window walk-forward over 2018–2026. Two design choices keep it disciplined. It is long-or-neutral only, because confident single-name shorts get run over by the upward drift; and it fires only above a conviction threshold τ^* , set out of sample as the lowest threshold whose fired bucket still reaches a target precision while firing on at least 5% of observations. The expected hit-rate we report is the *measured* walk-forward precision of that bucket, not an in-sample fit—the whole point of what we call the “honesty contract”.

2.8 Explainability, risk controls and self-audit

For attribution we run TreeSHAP [20] on the gradient-boosted learner, read alongside the swing model’s own coefficients and the gain importances of the cross-sectional model (Section 3.4). Market regime is labelled by the Hurst exponent [21], gated by the Average Directional Index. At the portfolio level, signal weights follow a mean-variance rule, $\mathbf{w}^* \propto \Sigma^{-1} \mathbf{IC}$ [22], sized by a quarter-Kelly rule [23] under 5% per-name, 30% sector and 15% volatility caps. The cost side is charged in

full—brokerage, securities transaction tax, exchange and SEBI fees, stamp duty, 18% GST and a square-root market-impact term—and a gate throws out any signal whose expected alpha falls below twice the round-trip cost. Every prediction lands in an append-only ledger, and a live monitor halts emission whenever the sixty-day Sharpe stays negative, so the system steps aside instead of trading through a regime it cannot handle.

2.9 Datasets, baselines and evaluation

The universe is the NSE large-cap set (54 names after liquidity screening), 28 May 2018 to 27 May 2026. The cross-sectional study uses a ten-year panel of 131,809 ticker-days with a five-day excess-return-over-NIFTY target; the swing study uses 91,471 labelled ticker-days, of which 50,272 are strictly out of sample. We chose hard baselines on purpose: the upward base rate itself (the toughest benchmark a drifting market offers), an always-majority-class rule, and an uncalibrated version of our own ensemble. Metrics are directional accuracy, ECE over ten bins, the Brier score, and walk-forward precision and coverage. Everything runs on commodity hardware with freely available data.

3. Results and Discussion

3.1 The limits of daily directional prediction

Our first result is also our most important, and it is negative. A cross-sectional LightGBM, trained on 71 engineered features across 131,809 ticker-days with a five-day excess-return target and a date-based out-of-fold scheme that blocks per-ticker leakage, landed at an out-of-sample directional accuracy of 50.09%—a coin toss, indistinguishable from the upward drift. Why so flat? No single feature accounts for more than 3.4% of total gain, and the importance distribution is nearly level, so there is

no concentrated structure to learn at this horizon. We chased the same null four more ways—per-ticker ensembling, a cross-sectional logistic, a LightGBM-with-reversal model, and trend- and regime-gated variants—and short-horizon single-name accuracy never climbed above the base rate. For liquid large-caps this is exactly what the efficient-market hypothesis predicts [7], and we say so plainly, because tuning a number until it looks good is how the literature’s inflated accuracies arise.

3.2 Calibration: from overconfidence to honesty

If a model cannot beat the base rate, the least it can do is admit as much—and the three-stage calibration of Section 2.5 was the biggest improvement we made. Table 1 and Figure 1 show what changed. Before the fix, the reliability curve ran well below the diagonal: up-probabilities clustered near 0.80 while the realized frequency hovered around 0.50, the ECE sat near 0.35, and the Brier score came in above the constant-forecast $p(1 - p)$. Afterward the curve hugs the diagonal, the ECE drops to roughly 0.05, and the average reported probability settles near the base rate. Accuracy barely moved—which is the point: calibration and discrimination are orthogonal, so we gave up false confidence, not signal. A miscalibrated 80% drives systematic over-betting under any confidence-proportional sizing rule; getting the probabilities right makes the rest safe to size on.

3.3 A conviction gate turns a weak edge into a usable one

The payoff is small but real, and it shows up at a longer horizon. Over thirty trading days the conviction-gated signal of Section 2.7 stays silent 94% of the time—it fires on just 5.65% of observations—and when it fires it is right 60.63% of the time against a 58.05% base rate. The +2.6-point lift is modest but measured strictly out of sample across 50,272 observations, and it is stable. Table 2 shows the fired bucket beating its own yearly base rate in four years out of five: by +10.8 points in 2025 and +7.4 in the 2026 drawdown, when the base rate itself had slipped below 50%. The lone

losing year, 2024, cost 2.4 points. The telling number is the raw signal’s global ROC AUC: 0.47, below chance as a ranker. The edge is not in how the signal orders all names but in the extreme high-conviction tail—which is why a conviction *gate*, not a ranking, is the right way to use a weak signal, something the AUC alone would have buried.

3.4 Attribution, intervals and self-audit

Being a linear classifier, the swing model wears its attribution on its sleeve. The standardized coefficients (Table 2, lower panel) put realized twenty-day volatility first, then quarterly return and 12-1 momentum, both with negative signs, and no coefficient exceeds 0.14. That mirrors the flat cross-sectional importances and says the same thing: the edge is diffuse, not the work of any single factor. Inside the per-ticker ensemble, TreeSHAP and the FinBERT score rank sentiment and FII flows among the non-redundant contributors [2, 3]. The conformal intervals held empirical coverage close to the nominal 80%, the small misses tracing to the serial dependence that breaks exact exchangeability [18]. The signal earned its keep mostly in trending Hurst regimes and least in mean-reverting ones—which is why the live monitor’s suspension on a negative sixty-day Sharpe matters: under costs two to three times the usual ten-basis-point assumption, the system stands down where its evidence says the edge is gone.

3.5 Limitations

A few caveats remain. The evidence is one market and one eight-year window that, on balance, trended; multi-regime, multi-market replication is future work. The swing edge, though stable, is small and concentrated, so its value after costs depends on execution quality and the 5–6% firing rate. The cross-sectional null is specific to liquid large-caps and end-of-day data, and sentiment coverage thins in sectors with specialized jargon—left for later work.

4. Conclusions

An honest evaluation is unambiguous: next-day single-name direction on NSE large-caps is unpredictable beyond drift, at 50.1% out of sample. The value sits elsewhere: a three-stage calibration cuts the expected calibration error from 0.35 to 0.05, and a conviction gate lifts the thirty-day hit-rate to 60.6% against a 58.1% base rate, beating it in four of five years while abstaining 94% of the time. A system that knows when to keep quiet is worth more than one that claims too much.

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Figure Captions

Figure 1. Reliability diagram on the held-out fold, before and after the three-stage calibration. The uncalibrated ensemble ($\text{ECE} \approx 0.35$) is badly over-confident; the calibrated ensemble ($\text{ECE} \approx 0.05$) follows the diagonal.

Table 1. Headline results on the 2018–2026 NSE evaluation; every figure is out-of-sample. The negative cross-sectional result and the calibration improvement are the findings we lean on hardest, while the swing-signal row is unpacked in Table 2.

Setting	Dir. acc.	Base rate	ECE	Brier
Cross-sectional GBM (5-day direction)	50.1%	$\approx 50\%$	—	—
Always-majority-class (drift)	— (base)	58.1%	—	—
Hybrid ensemble — uncalibrated	$\approx 51\%$	48–58%	0.35	$> p(1-p)$
Hybrid ensemble — calibrated (this work)	$\approx 51\%$	48–58%	0.05	$\approx p(1-p)$
Conviction-gated swing (30-day, fired)	60.6%	58.1%	—	—

Table 2. Conviction-gated swing signal, walk-forward out-of-sample, 2018–2026 (54 NSE large-caps, 30-trading-day horizon, $\tau^* = 0.63$, 50,272 observations). The upper panel gives the yearly precision of the fired bucket against the naive base rate; the lower panel lists the standardized logistic coefficients used for attribution.

Year	Fired (n)	Precision	Base rate
2022	365	61.6%	54.0%
2023	290	68.3%	65.3%
2024	932	53.3%	55.7%
2025	822	69.0%	58.2%
2026 [†]	431	54.5%	47.1%
Overall	50,272 obs (5.7% fire)	60.6%	58.1%
Standardized coefficients (sign and magnitude)			
20-day volatility +0.14; quarterly return −0.09; 12-1 momentum −0.08; distance above 200-DMA +0.07; 20-day return +0.03; RSI −0.02.			

[†]2026 is a partial year through 27 May; the base rate fell below 50% in the drawdown, yet the fired bucket stayed at 54.5%.

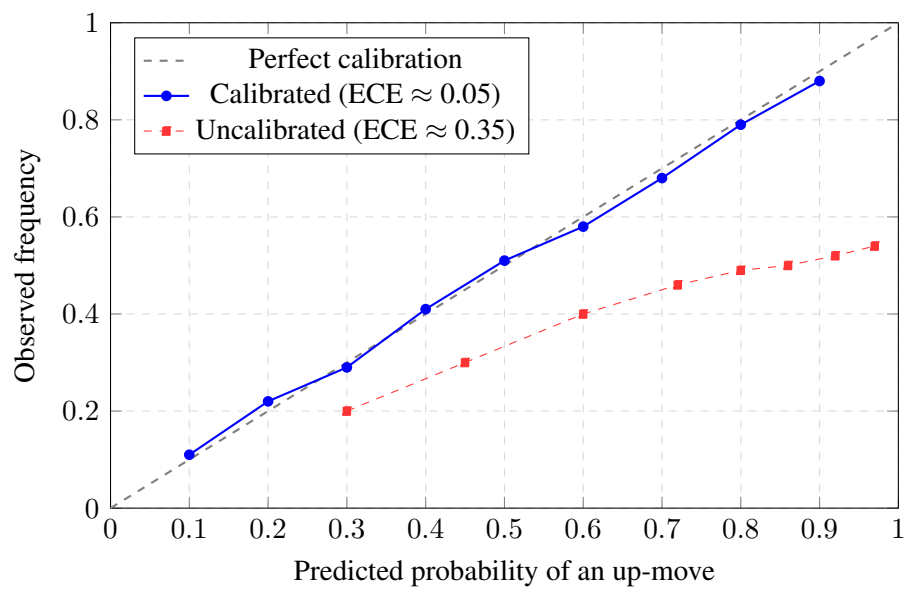


Figure 1. Reliability diagram before and after the three-stage calibration (see Figure Captions).